DORA BABU CHINTALA

Falls Church, VA | +1 (909) 802-6515 | chintala.d23@gmail.com | Portfolio

PROFESSIONAL SUMMARY

- Senior Software Engineer with 8+ years of experience building enterprise applications and AI-enabled software solutions using Java, Spring Boot, Angular, AWS, microservices, and distributed architectures.
- Specializes in Generative AI and Agentic AI application development using LLMs, RAG, LangChain, MCP, Prompt Engineering, Vector Databases, Knowledge Retrieval, and LLM Integration.
- Combines full-stack and cloud engineering with AI capabilities to modernize legacy applications, analyze system dependencies, automate knowledge discovery, and deliver context-aware intelligent workflows.
- Experienced across AWS EKS, Kubernetes, Docker, Kafka, REST APIs, CI/CD, observability, testing, and system design, supporting scalable software from architecture and development through production delivery.

TECHNICAL SKILLS

Programming Languages: Java 17/21, Python, TypeScript, JavaScript, SQL
Backend Engineering: Spring Boot, Spring Security, Spring Batch, Hibernate, JDBC, REST APIs
Frontend Engineering: Angular 19, REST API Integration
Distributed Systems & Architecture: Apache Kafka, Event-Driven Architecture, Distributed Systems, Microservices Architecture, API Design, Design Patterns, Service Integration
Cloud & Containers: AWS, Amazon EKS, Kubernetes, Docker, Cloud-Native Applications
Databases & Persistence: MongoDB, ORM, Transaction Management
Build & DevOps: Gradle, Jenkins, Git, CI/CD, Sonar, Dependency Management, Build Automation
Testing: JUnit, Cucumber, Cypress, Unit Testing, Integration Testing, Regression Testing, UI Automation
Observability & Reliability: ELK Stack, Prometheus, Grafana, Distributed Tracing, Logging, Metrics, Production Troubleshooting, Resiliency
Generative AI & Agentic AI: Generative AI (GenAI), Agentic AI, LLMs, RAG, LangChain, MCP, Prompt Engineering, Vector Databases, Knowledge Retrieval, LLM Integration
Software Design & Delivery: System Design, High-Level Design (HLD), Low-Level Design (LLD), Framework Modernization, Legacy Modernization, SDLC, Code Reviews

PROFESSIONAL EXPERIENCE

Senior Software Engineer **Oct 2024 - Present**
Freddie Mac | McLean, VA

- Applied LLM-driven knowledge retrieval and Agentic AI techniques alongside Java and Spring Boot analysis across 30+ legacy batch jobs, helping engineering teams identify dependencies and business rules for phased application modernization.
- Designed event-driven loan-processing workflows with Spring Boot, Apache Kafka, and Microservices Architecture across 12+ distributed services, increasing processing consistency while limiting tight dependencies between downstream systems.
- Scaled Java, Kafka, Hibernate, JDBC, and SQL processing to handle approximately 40K loan records per cycle, strengthening transaction integrity and giving operations teams more dependable reconciliation outcomes.
- Accelerated regression validation by 75% through Cypress, Angular 19, Jenkins, and CI/CD automation, giving developers and QA teams faster feedback before mortgage application changes reached production.
- Delivered cloud-native mortgage capabilities with Java, Spring Boot, Angular 19, REST APIs, AWS EKS, Kubernetes, and Docker, connecting frontend workflows with scalable backend services for dependable production processing.
- Integrated Agentic AI and RAG workflows using LLMs, LangChain, MCP, Prompt Engineering, and Knowledge Retrieval to analyze legacy application dependencies, extract business rules, and reduce manual investigation during modernization.
- Strengthened release controls through Gradle, Git, Sonar, JUnit, Cucumber, Dependency Management, and Code Reviews, enabling development, QA, and platform teams to catch compatibility risks before deployment.
- Stabilized production services with ELK Stack, Prometheus, Grafana, Distributed Tracing, Logging, and Metrics, giving support teams faster visibility into failures and sustaining reliable mortgage-processing operations.

Software Engineer **Jun 2023 - Sep 2024**
Fidelity Investments | Westlake, TX

- Engineered Spring Boot services with Spring Security, Hibernate, SQL, and REST APIs for financial applications, giving business systems secure and maintainable access to backend transaction capabilities.
- Deployed Dockerized microservices on AWS EKS and Kubernetes with platform teams, establishing repeatable release practices that supported zero-downtime delivery across integrated application services.
- Established reusable REST integrations across 12+ services using Spring Boot, Hibernate, and API Design, centralizing common backend behavior and decreasing duplicated integration work across financial systems.
- Automated recurring financial-processing workflows with Spring Batch, Java, SQL, and Transaction Management, giving operations teams predictable execution while preserving data integrity across downstream services.
- Connected financial applications through Apache Kafka and Event-Driven Architecture, enabling asynchronous service integration and resilient message processing while limiting direct dependencies between distributed services.
- Secured internal APIs with Spring Security and REST controls, partnering with development and QA teams to validate access paths and protect service-to-service communication in production.
- Upgraded application frameworks through Gradle, Git, Jenkins, Sonar, CI/CD, and Code Reviews, helping engineering teams identify compatibility issues earlier and move dependency changes through releases with lower operational risk.
- Diagnosed production issues across Java services, Kafka messaging, SQL persistence, and Kubernetes workloads using Logging, Metrics, and Production Troubleshooting, restoring transaction flows and protecting continuity across dependent financial applications.

Graduate Assistant - Software Engineering **Sep 2021 - Jun 2023**
University of Illinois Chicago | Chicago, IL

- Developed Java and Spring Boot applications for university initiatives, converting functional requirements into REST APIs and SQL-backed workflows that gave faculty and staff dependable access to operational functions.
- Implemented backend services with Spring Boot, REST APIs, and SQL, creating cleaner application-to-data interactions that made university systems easier for technical contributors to maintain and extend.
- Optimized Java service logic and SQL access patterns to improve application response time by 30%, giving university users faster request handling across supported academic and administrative workflows.
- Automated data-processing routines with Python and SQL, increasing processing efficiency by 25% while preserving data accuracy across academic and administrative workflows.
- Structured reusable Java and Spring Boot components with Design Patterns and object-oriented principles, separating service responsibilities so developers could extend university applications with less rework and lower regression risk.
- Corrected backend issues by tracing Java execution, REST API behavior, and SQL queries, restoring expected application behavior and limiting interruptions to daily university workflows.
- Verified service and database changes through JUnit, Integration Testing, and Regression Testing, catching defects before release and increasing confidence in updates delivered to university users.
- Documented REST API behavior, application flows, and implementation decisions with university stakeholders, giving technical contributors clearer support guidance and simplifying future maintenance and handoff.

Software Engineer - Full Stack **May 2018 - Aug 2021**
Aadyam Solutions Pvt. Ltd. | India

- Created full-stack web features with Java, Spring Boot, Angular, TypeScript, REST APIs, and SQL, translating business requirements into working functionality across frontend, backend, and persistence layers.
- Assembled Angular interfaces with REST API Integration, connecting user actions to Spring Boot services so application workflows behaved consistently from the browser through backend processing.
- Implemented persistence flows with Hibernate, JDBC, SQL, ORM, and Transaction Management, supporting reliable create, update, and retrieval operations across core enterprise application functions.
- Connected Spring Boot components through Apache Kafka, enabling asynchronous event handling and giving application modules more flexibility than tightly coupled request-response interactions.
- Standardized build and release workflows with Gradle, Git, Jenkins, CI/CD, Dependency Management, and Build Automation, helping developers package and promote application changes consistently across delivery environments.
- Refactored Java and Spring Boot components with Design Patterns, simplifying service logic and making new application features easier to add without disrupting established functionality.
- Validated frontend and backend changes through Integration Testing, Regression Testing, and UI Automation, catching cross-layer defects earlier and supporting more dependable application releases.
- Troubleshot Angular screens, REST APIs, Java services, and SQL persistence through Logging and Production Troubleshooting, restoring expected application behavior and giving delivery teams more stable releases.

PROJECTS

AI-Enabled Healthcare Professional Portal

- Built an AI-enabled healthcare portal with Angular 19, Spring Boot, REST APIs, and MongoDB across 4 core workflows covering professional profiles, credential verification, employer matching, and onboarding.
- Structured a 5-layer architecture spanning frontend, backend, persistence, AI, and retrieval using System Design, HLD, LLD, and API Design, enabling independently maintainable services and scalable platform expansion.
- Developed a RAG-based learning capability using LangChain, LLM Integration, Prompt Engineering, Vector Databases, and Knowledge Retrieval to ground personalized assessments and study guidance in relevant learner context.

EDUCATION

Master of Science, Management Information Systems **2023**
University of Illinois Chicago | Chicago, IL

Bachelor of Technology, Mechanical Engineering **2018**
GITAM University | India

CERTIFICATIONS

- AWS Academy - **Data Analytics**
- AWS Academy - **Cloud Foundations**
- AWS Academy - **Cloud Security Foundations**
- IBM Full Stack Software Developer Professional Certificate - **Coursera**
- Meta Back-End Developer Professional Certificate - **Coursera**
- Spring: Spring Boot - **LinkedIn Learning**